

racks. With most equipment running auto-ranging power supplies, many data centres have moved to 20A or 30A, 220V (more efficient) redundant feeds for their racks. Each 220V circuit requires a two-pole circuit breaker, effectively reducing the number of racks that can be supported from 42 to 21, even if the power draw going to the racks is less than the power available.

With blade servers running on 3-phase power, circuit breaker management becomes difficult. Each blade server chassis typically requires redundant 3-phase connections. Put six chassis in a rack and that is up to twelve 3-phase connections. Just as the term 3-phase implies, it takes three circuit breaker positions in a panel. So a fully loaded rack of blade chassis requires up to 36 circuit breaker positions. This leaves just six positions open in a 42-breaker panel.

Many blade chassis manufacturers have realised this problem and developed a rack power distribution system to reduce the number of circuit breakers needed to support a rack of blade chassis. This will reduce a typical 42-circuit-breaker panel to supporting 14 racks, assuming that there is enough power feeding the panel.

Green data centre

A green data centre is a repository for the storage, management and dissemination of data in which the mechanical, lighting, electrical and computer systems are designed for maximum energy efficiency and minimal environmental impact. The construction and operation of a green data centre includes advanced technologies and strategies. Here are some examples:

1. Minimising the footprints of the buildings
2. Use of low-emission building materials, carpets and paints
3. Sustainable landscaping
4. Waste recycling
5. Installation of catalytic converters on backup generators
6. Use of alternative energy technologies like photovoltaic, heat pumps

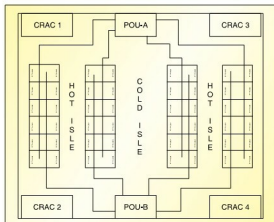


Fig. 3: Distributed bus layout

- and evaporative cooling
7. Use of hybrid or electric company vehicles

Building and certifying a green data centre or other facility can be expensive up front, but long-term cost savings can be realised on operations and maintenance. Another advantage is that green facilities offer employees a healthy, comfortable work environment. In addition, these enhance relationships with local communities.

There is growing pressure from environmentalists and the general public, for governments to offer green incentives like monetary support for the creation and maintenance of ecologically responsible technologies.

There are several interesting options for conserving energy in today's office environment. Data centres generate heat all year round, and cooling this heat requires air-conditioners and various methods of heat rejection. During winters, in regions that require normal office heat, there are methods of using outside cold air and/or water to remove the heat. A better use of that heat would be to supplement the heating needs of the non-data centre areas of the building—areas where people work or that require heating and climate control for other reasons.

One of the difficulties in maintaining a clean data centre is the quality of the air. It appears that the simplest solution for climates that experience cold winter temperatures would be to open

the window and let the hot air out and replace it with cold air. The problem with this solution is that you rapidly lose control of the air quality. Air quality, in this case, refers to the amount of dirt and humidity. You could filter the air, but it requires energy to force the air through the filters and frequent filter changes as the dirt from outside clogs the filters.

During the cold winter months, the outside hu-

midity is quite low and a significant amount of humidification needs to be provided to the data centre to compensate for the low humidity in the outside air. Also, keep in mind that while the data centre is running air-conditioning, the rest of the building is typically being heated. Rather than throwing away the heat from the data centre, investigate ways to use the heat energy from the data centre to heat other parts of the building.

Various air-to-air and liquid heat exchangers are available that supplement the building's heating system with the heat generated by the data centre. This type of system can also be used in hot summer months with outside air rather than air from the populated space of the building. Even on a 90-degree day, the air temperature in the hot-air return space is likely to be 10–30 degrees hotter.

Structured data cabling. To provide a flexible solution, you must define criteria for the data centre, such as the heat load and power that will be allowed on a per rack basis. This helps to define how many servers will be permitted in a rack. Following the number of servers is the number of network (LAN and SAN) connections required.

Design areas of the data centre that will be used for SAN and network consolidation points. This normally requires a careful LAN and SAN design process to identify current and future technologies that will be supported by the envi-